

Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia



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Background and Objectives



The *Bacillus cereus* group is a diverse group of spore-forming bacteria classified into 8 *panC* phylogenetic groups. **Some *B. cereus* group members classified as biovar Thuringiensis are used as bioinsecticides.**



In 2022, European Union member states reported 127 foodborne outbreaks attributed to *Bacillus cereus* toxins¹. Although foodborne illness attributed to *B. cereus* species is typically associated with *B. cereus sensu stricto* (s.s.) and biovar Emeticus, the **recovery of biovar Thuringiensis from clinical cases has raised concerns².**



Current food safety frameworks often rely on taxonomic identification alone, which can overlook important differences in virulence potential. By focusing on isolates recovered from human clinical cases, **which are underrepresented in existing genomic and cytotoxicity datasets**, this work strengthens our understanding of strain characteristics most relevant to human clinical cases³.

Objective

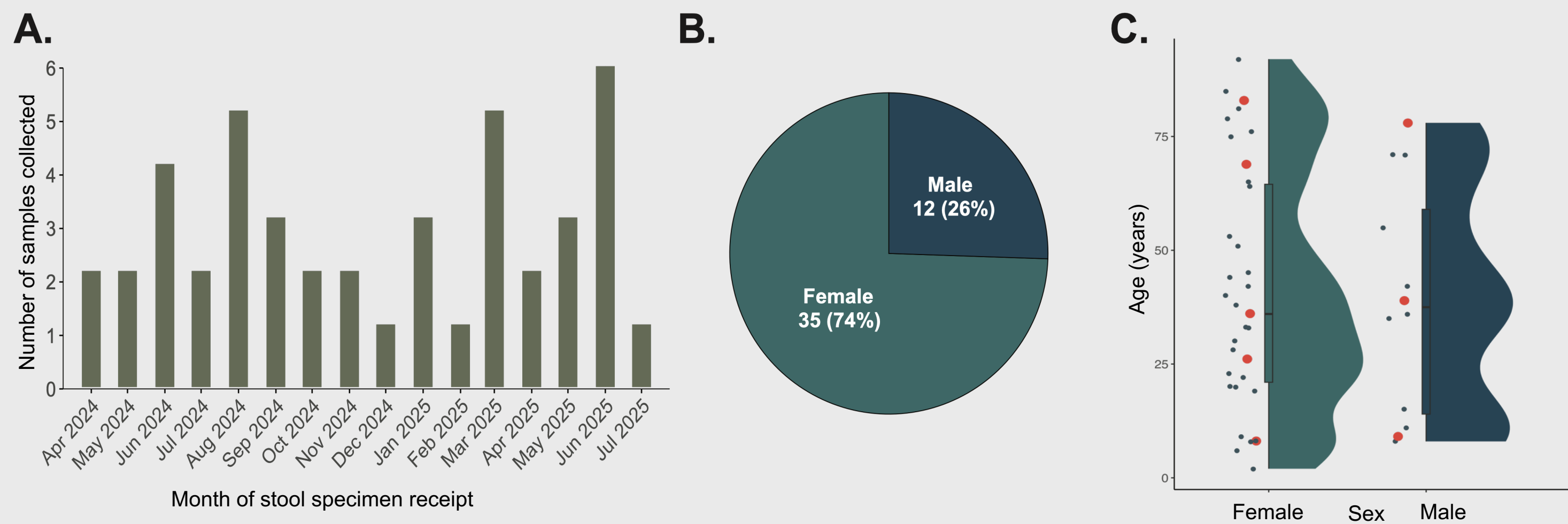
Characterize the genotypic and phenotypic diversity of 47 *B. cereus* group isolates recovered from human clinical cases in Slovenia

Significance

Characterizing *B. cereus* isolates from human clinical samples can strengthen our understanding of strain characteristics linked to human illness

1 When and who were *B. cereus* isolates collected from?

Figure 1. *B. cereus* isolates were recovered from human clinical cases by the Slovenian National Laboratory of Health, Environment, and Food between April 2024 and July 2025. **(A)** Temporal distribution of isolation dates showing collection months of *B. cereus* from stool samples. **(B)** Sex distribution of cases, with females representing 74% (35/47) and males 26% (12/47) of isolates collected. **(C)** Age distribution stratified by sex, displayed as a raincloud plot showing individual data points (colored by hospital admission status: red = hospitalized, blue = primary care or outpatient), density distributions, and boxplot summaries.



2 What *B. cereus* group species are present in human clinical samples and do they carry virulence genes?

Methods

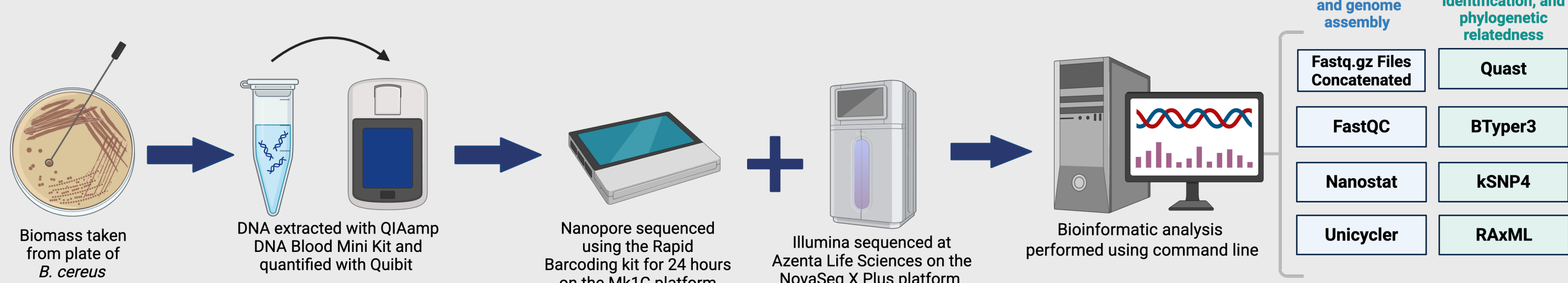


Figure 2. Both Nanopore and Illumina sequencing was conducted to create hybrid assemblies. Read quality, genome assembly, and genome quality assessment was performed using FastQC (v 0.12.1) along with Nanostat (v 1.4.0), Unicycler (v 0.5.1) and QUAST (v 5.2.0), respectively. Taxonomic classification and virulence gene identification were performed with BType3 (v 3.4.0), and phylogenetic grouping with kSNP4 (v 4.1.0). RAxML (v 2.0.2) was used to construct a maximum-likelihood phylogenetic tree, which was visualized in the interactive tree of life (iTOL) (v6) software.

Results

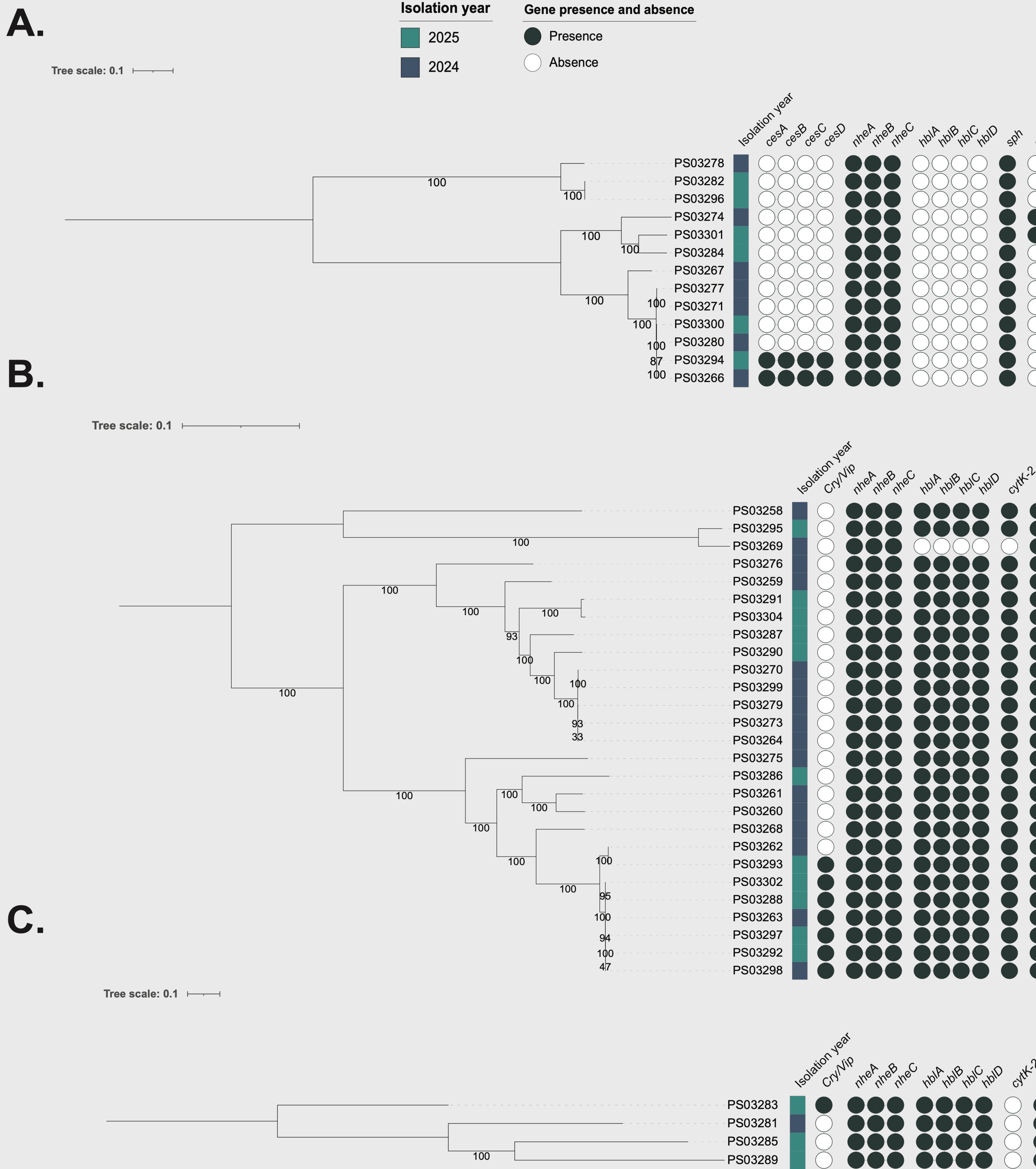


Figure 3. Maximum-likelihood phylogenetic tree of the *B. cereus* group isolates recovered from human clinical samples. Isolates were classified as **(A)** *B. mosaicus* (n=13, including 2 biovar Emeticus), **(B)** *B. cereus* s. s. (n=27, including 7 biovar Thuringiensis), **(C)** *B. toyonensis* (n=4, including 1 biovar Thuringiensis), and *B. mosaicus/luti* (Not shown, n=3). BType3 detected *nhe* genes in 100%, *hbl* in 63%, *cytK-2* in 59%, and *sph* in 100% of isolates, while *cesABCD* was present in two isolates (4%)⁴.

3 Do *B. cereus* isolates recovered from human clinical samples exhibit toxicity toward human cells?

Methods

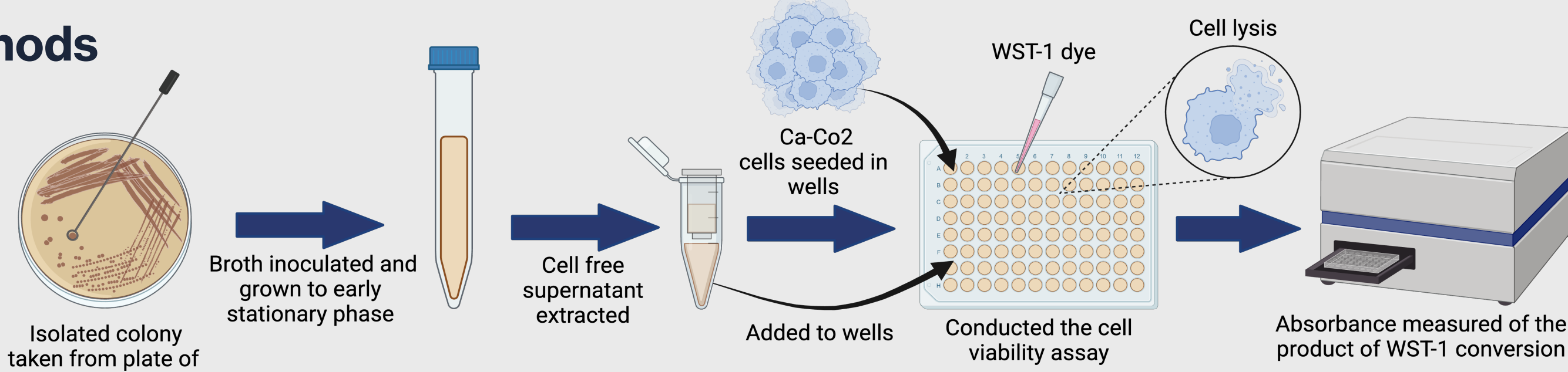


Figure 4. Isolates were grown in brain heart infusion (BHI) broth to early stationary phase at 37 °C to collect cell-free supernatants. Cytotoxicity of supernatants was assessed using the WST-1 assay on Caco-2 cells. Caco-2 cells were exposed to 15% v/v of supernatants for 15 minutes before adding the WST-1 dye. Absorbance values were min-max normalized to BHI and *B. cereus* ATCC 14579, respectively, to determine cytotoxicity.

Results

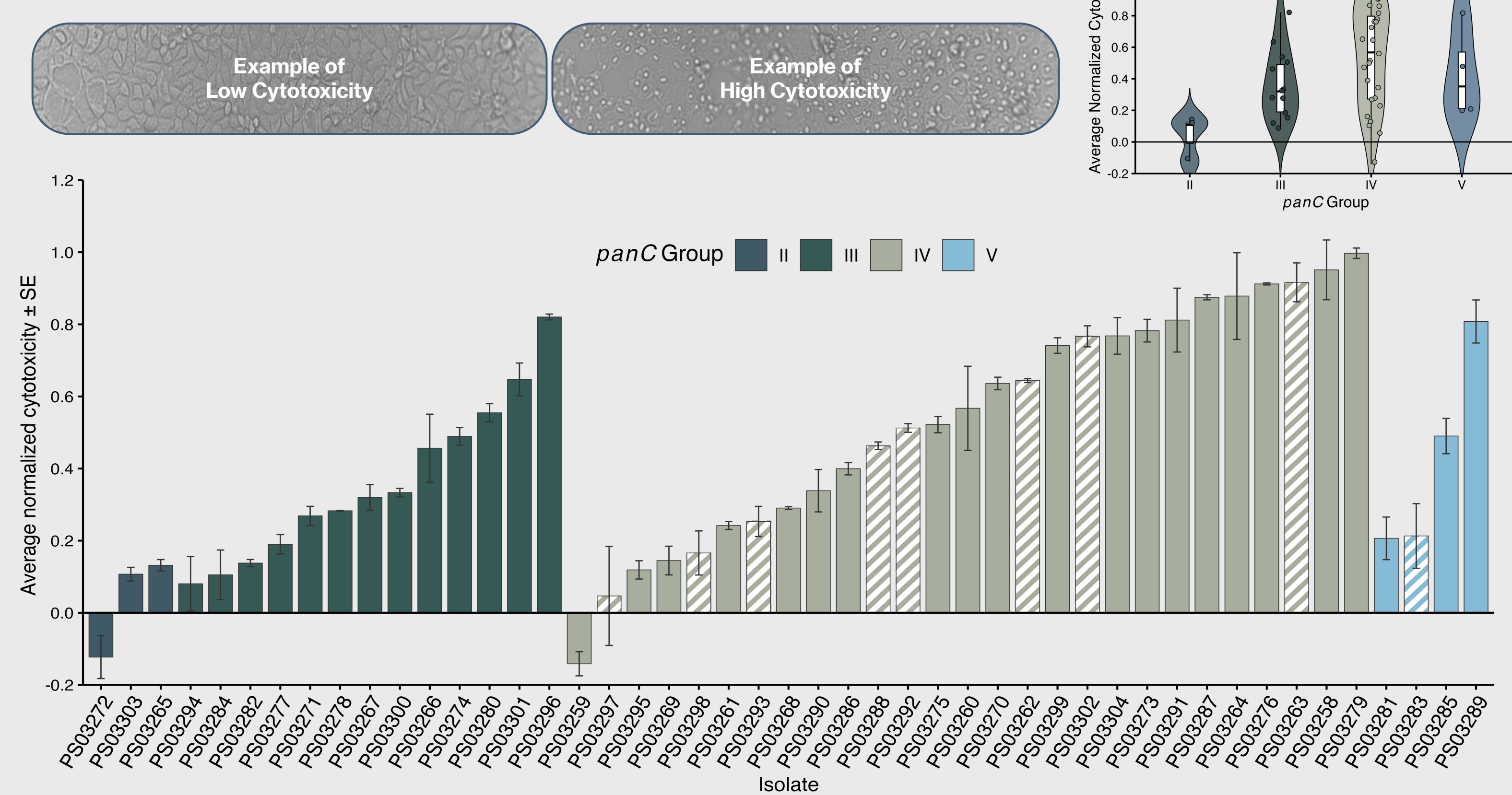


Figure 5. Data represents the average of two experiments, each with six technical replicates, and error bars indicate the standard deviation. **Isolates with striped lines were identified as biovar Thuringiensis.** Isolates from group IV (*B. cereus* s.s.) had the highest average cytotoxicity (0.57 ± 0.32). In contrast, group II showed the lowest cytotoxicity (0.119 ± 0.02), with significant differences between *panC* groups ($p = 0.0187$).

4 Are biovar Thuringiensis isolates genomically similar to commercial biopesticide strains?

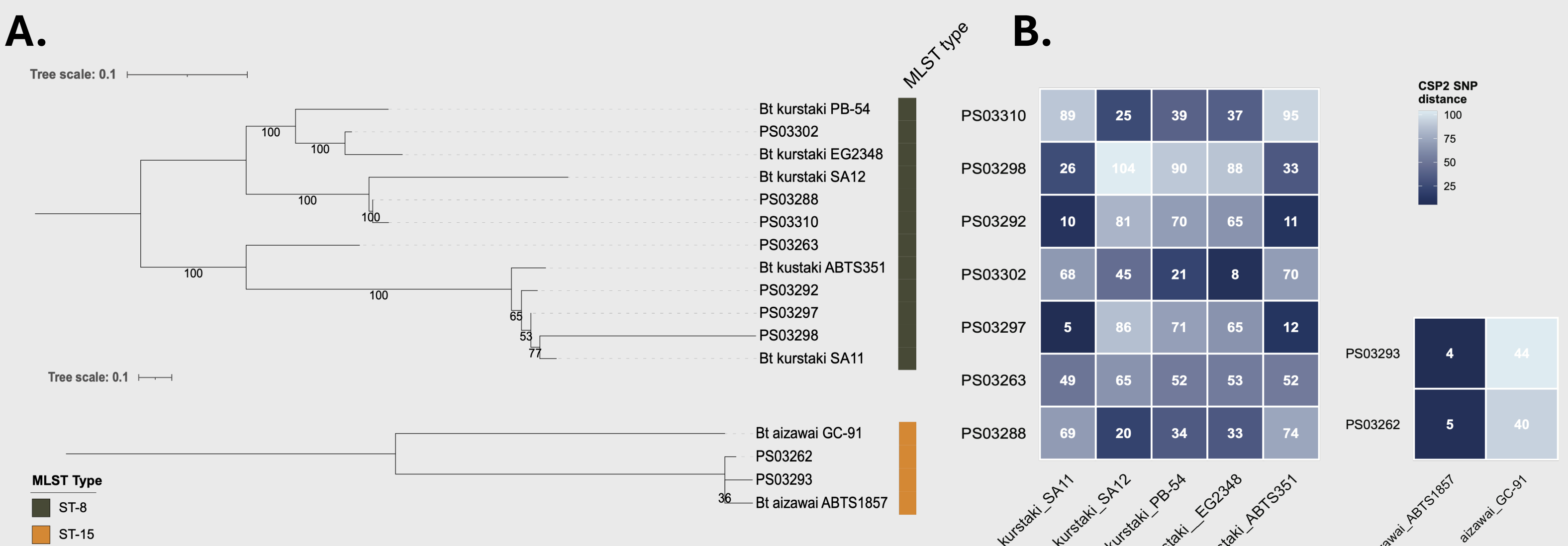


Figure 6. **(A)** Maximum-likelihood phylogenetic tree of *B. cereus* s.s. biovar Thuringiensis isolates compared to commercial bioinsecticide strains using Illumina only assemblies. Isolates are color-coded by MLST sequence type: ST-8 (green) represents the *kurstaki*-associated cluster, and ST-15 (orange) represents the *aizawai*-associated cluster. **(B)** SNP analysis using the CSP2 SNP pipeline (v 0.9.7) revealed 5-104 filtered SNPs within the ST-8 group relative to *kurstaki* strains, and 5-44 filtered SNPs within the ST-15 group relative to *aizawai* strains.

Conclusions

1 *B. cereus* s.s. predominated among clinical isolates, with seven biovar Thuringiensis isolates.

The high genomic relatedness of biovar Thuringiensis isolates to commercial biopesticide strains underscores the need for exposure assessment frameworks that integrate genomic and phenotypic data to more accurately predict food safety risk associated with these strains.

Future Directions



Validate a SNP-based cytotoxicity prediction model using the genomic and phenotypic data from this clinical isolate dataset.



Strengthen existing exposure assessment models with phenotypic data collected in this study to better predict which *B. cereus* strains are more likely to cause illness.

References

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- (2) Biggel, M., N. Jessberger, J. Kovac, and S. Johler. 2022. Recent paradigm shifts in the perception of the role of *Bacillus thuringiensis* in foodborne disease. *Food Microbiology* 105:104025.
- (3) Carroll, L. M., R. A. Cheng, M. Wiedmann, and J. Kovac. 2022. Keeping up with the *Bacillus cereus* group: taxonomy through the genomics era and beyond. *Critical Reviews in Food Science and Nutrition*. Taylor & Francis 62:7677–7702.
- (4) Carroll, L. M., R. A. Cheng, and J. Kovac. 2020. No Assembly Required: Using BType3 to Assess the Congruency of a Proposed Taxonomic Framework for the *Bacillus cereus* Group With Historical Typing Methods. *Frontiers in Microbiology* 11:580691.